↑ AFFIX SEAL HERE ↑

CANDIDATE – PLEASE NOTE!

You must sign below and return this booklet with the Answer Sheet. Failure to do so may result in disqualification.

____
Signature

FORM TP 22137

TEST CODE **002401**

MAY/JUNE 2002

CARIBBEAN EXAMINATIONS COUNCIL

SECONDARY EDUCATION CERTIFICATE EXAMINATION

PHYSICS

Paper 01 – General Proficiency

$1\frac{1}{4}$ hours

(10 JUNE 2002 (a.m.))

READ THE FOLLOWING DIRECTIONS CAREFULLY

1. In addition to this test booklet, you should have an answer sheet.
2. Each item in this test has four suggested answers lettered (A), (B), (C), (D). Read each item you are about to answer and decide which choice is best.
3. On your answer sheet, find the number which corresponds to your item and shade the space having the same letter as the answer you have chosen. Look at the sample item below.

Sample Item

The SI unit of length is the

(A) newton
(B) metre
(C) kilogram
(D) second

Sample Answer

(A) ● (C) (D)

The best answer to this item is "metre" so answer space (B) has been shaded.

4. If you want to change your answer, be sure to erase your old answer completely and fill in your new choice.
5. When you are told to begin, turn the page and work as quickly and as carefully as you can. If you cannot answer an item, omit it and go on to the next one. You can come back to the harder item later. Your score will be the total number of correct answers.
6. Figures are not necessarily drawn to scale.
7. You may do any rough work in this booklet.
8. The use of non-programmable calculators is allowed.
9. This test consists of 60 items. You will have 75 minutes to answer them.
10. Do not be concerned that the answer sheet provides spaces for more answers than there are items in this test.

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DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

2002

1. Which of the following physical quantities does NOT have a unit?

(A) Refractive index
(B) e.m.f.
(C) Density
(D) Frequency

2. The base unit of temperature of the S.I. system is the

(A) degree Celsius
(B) degree Fahrenheit
(C) Kelvin
(D) degree centigrade

3. The mass of a piece of metal is 43.7 g and its volume is 5.6 cm³. A student uses a calculator to find the density of the metal and obtains the figure 7.803571. He should write the density as

(A) 7.803571 g cm^{-3}
(B) 7.80 g cm^{-3}
(C) 7.8 g cm^{-3}
(D) 8.0 g

4. A simple pendulum oscillates about centre O between positions X and Y. Which of the following could represent one complete oscillation? (The arrows indicate the direction of movement.)

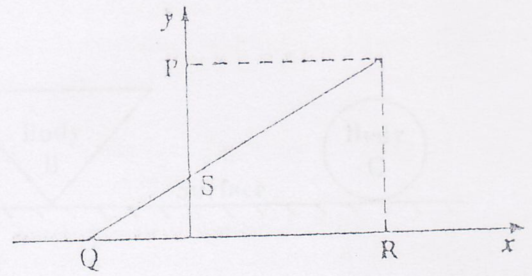
I. X → O → Y → O → X
II. O → X → O → Y → O
III. Y → O → X

(A) I only
(B) III only
(C) I and II only
(D) I, II and III

5. The unit of electrical resistance may be expressed as

(A) 1Ω = 1 V.A^{-1}
(B) 1Ω = 1 A.V
(C) 1Ω + 1 A.V^{-1}
(D) 1Ω = 1 W.A^{-1}

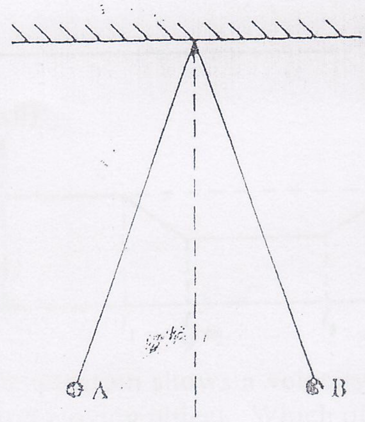
6. Item 6 refers to the diagram below.



When $x = 0$, the value of y is

(A) Q
(B) S
(C) P
(D) P/R

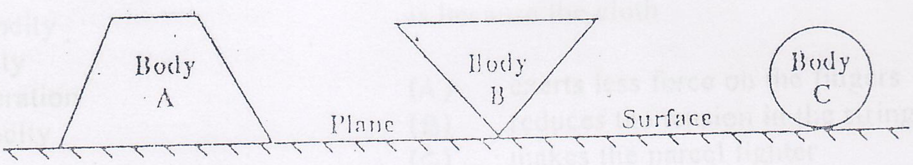
7.



Two light, styrofoam spheres, A and B, are suspended by insulating threads. If they are at rest in the positions shown in the diagram, the force keeping them apart is

(A) gravitational force
(B) electrostatic force
(C) magnetic force
(D) centripetal force

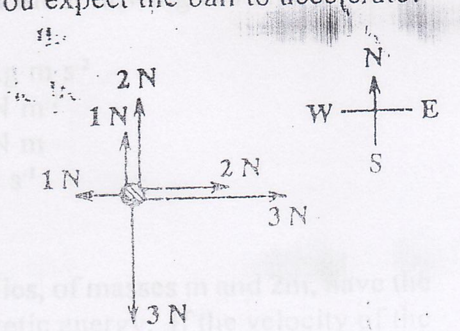
8. In relation to the diagrams below, which of the following statements is/are correct?



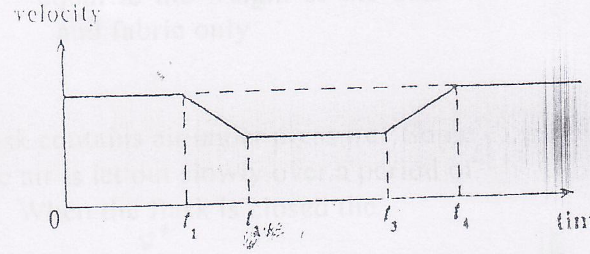
I. Body A is in neutral equilibrium.
II. Body B is in unstable equilibrium.
III. Body C is in stable equilibrium.

(A) II only
(B) I and II only
(C) II and III only
(D) I, II and III

9. Six forces act on a ball as shown in the diagram below. In which direction would you expect the ball to accelerate?



(A) North
(B) South
(C) East
(D) West

10. 

The diagram shows a velocity/time graph for a moving object. Which of the following statements about the object would be true?

I. It returns to its starting point.
II. It has zero acceleration between times t_2 and t_3.
III. Its velocity at t_4 is the same as its initial velocity.

(A) II only
(B) I and II only
(C) II and III only
(D) I, II and III

11. If the resultant force on an object is zero, the object can move with

(A) decreasing velocity
(B) constant velocity
(C) constant acceleration
(D) increasing velocity

12. The constant unbalanced force acting on a body is at right angles to its velocity. Which of the following statements about this motion is correct?

(A) The body has no acceleration.
(B) The kinetic energy of the body is increasing.
(C) The body is slowing down.
(D) The body is moving in a circular path.

13 Which of the following is a unit of work?

(A) kg m s^{-2}
(B) N m^{-2}
(C) N m
(D) J s^{-1}

14 Two bodies, of masses m and 2m, have the some kinetic energy. If the velocity of the smaller body is v, what is the velocity of the other body?

(A) $\frac{v}{\sqrt{2}}$

(B) $\frac{v}{2}$

(C) $\sqrt{2}\,v$

(D) $2\,v$

15. A heavy parcel is tied with string. It is less painful for a person to pick up the parcel if a cloth is wrapped around the string. This is because the cloth

(A) exerts less force on the fingers
(B) reduces the tension in the string
(C) makes the parcel lighter
(D) reduces the pressure on the fingers

16. A balloon filled with hydrogen gas and released accelerates upwards. The balloon must therefore have displaced a weight of air

(A) less than its own weight
(B) equal to its own weight
(C) greater than its own weight
(D) equal to the weight of the basket and fabric only

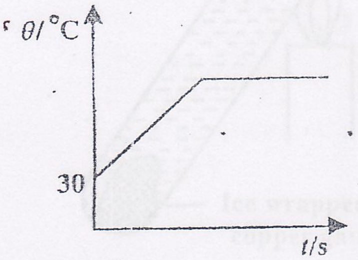
17. A flask contains air under pressure. Some of the air is let out slowly over a period of 10 s. When the flask is closed the

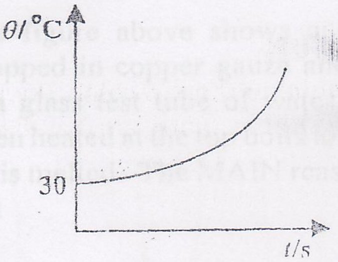
(A) pressure of the air in the flask will have increased
(B) volume of air in the flask will have decreased
(C) temperature in the flask will have increased
(D) number of molecules striking the wall per second will have decreased

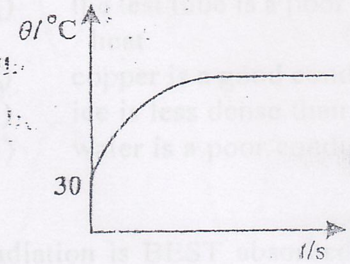
18. The specific heat capacity of a material is the energy required to

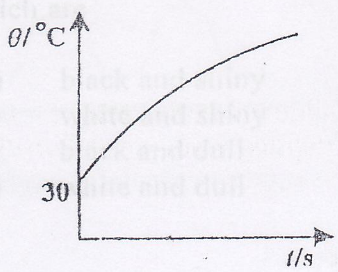
(A) melt 1 kg of the material with no change of temperature
(B) change the temperature of the material by 1 K
(C) change 1 kg of the liquid material to 1 kg of gas without a change in temperature
(D) change the temperature of 1 kg of the material by 1 K

19. An uninsulated closed container with water is heated from room temperature (30°C) until all the water has evaporated. Which of the following graphs most accurately represents the variation of water temperature (θ) with time (t)?

(A)


(B)


(C)


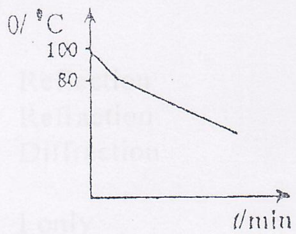
(D)


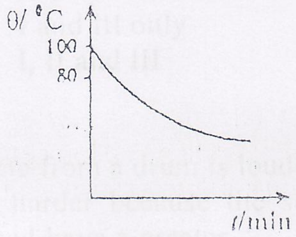
20. Which of the following would be responsible for the heating of the Earth by the Sun?

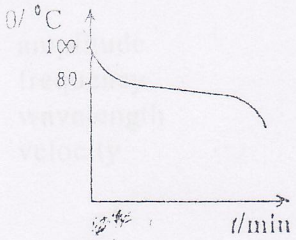
I. Radiation
II. Convection
III. Conduction

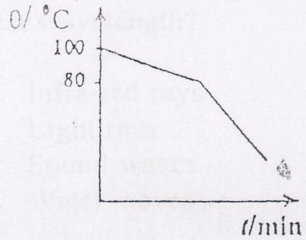
(A) I only
(B) III only
(C) I and II only
(D) II and III only

21. Some molten naphthalene at 100°C is allowed to cool down at room temperature. If naphthalene has a melting point of 80°C, which of the following BEST represents the cooling curve?

(A)


(B)


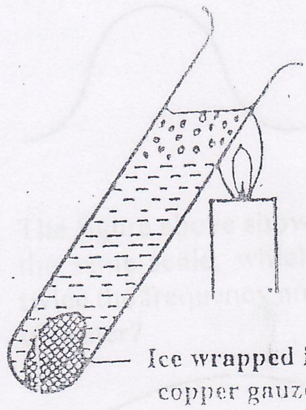
(C)


(D)


22. An experiment is set up with a smoke cell for demonstrating Brownian motion. The moving specks observed are

(A) smoke particles seen by reflected light
(B) smoke particles in constant vibration
(C) molecules of vibrating air
(D) molecules of carbon in random

23.



The figure above shows a piece of ice wrapped in copper gauze and submerged in a glass test tube of water. The water when heated at the top boils long before the ice is melted. The MAIN reason for this is that

(A) the test tube is a poor conductor of heat
(B) copper is a good conductor of heat
(C) ice is less dense than water
(D) water is a poor conductor of heat

24. Radiation is BEST absorbed by surfaces which are

(A) black and shiny
(B) white and shiny
(C) black and dull
(D) white and dull

25. According to the kinetic theory, when a gas in a closed container is heated the pressure rises because

(A) there are more molecules hitting the walls of the container
(B) the molecules move faster and hit each other harder and more often
(C) the molecules move faster and hit the walls of the container with greater force and with greater frequency
(D) the molecules expand and push harder on the walls of the con-

26. Water waves which are produced in a ripple tank travel more slowly as they move from deep to shallow water. Which of the following can this fact be used to demonstrate?

I. Reflection
II. Refraction
III. Diffraction

(A) I only
(B) II only
(C) I and III only
(D) I, II and III

27. The note from a drum is louder when it is struck harder because the sound waves produced have a greater

(A) amplitude
(B) frequency
(C) wavelength
(D) velocity

28. Which of the following would have the shortest wavelength?

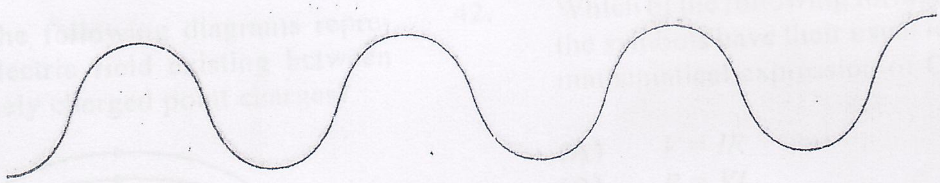
(A) Infra-red rays
(B) Light rays
(C) Sound waves
(D) Water waves

29. Which of the following statements is/are true?

I. Thermal radiation is a type of electromagnetic radiation.
II. Blue light has a longer wavelength than infra-red radiation.
III. The velocity of X-rays in a vacuum is 300 000 000 m s^{-1}.

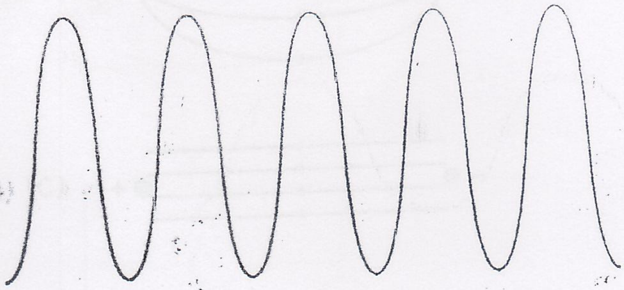
(A) I and II only
(B) I and III only
(C) II and III only
(D) I, II and III

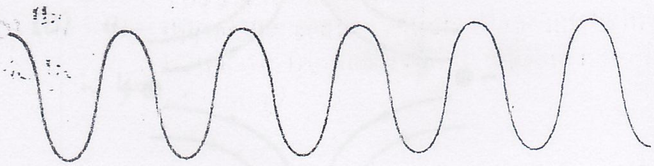
30.



The figure above shows the profile of a water wave. Using the same scale, which diagram below represents a wave twice the frequency and half the amplitude in the same tank of water?

(A) 

(B) 

(C) 

(D) 

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31. The sharp edges of shadows suggest that light

(A) has a wave nature
(B) is a form of energy
(C) travels in straight lines
(D) travels very quickly

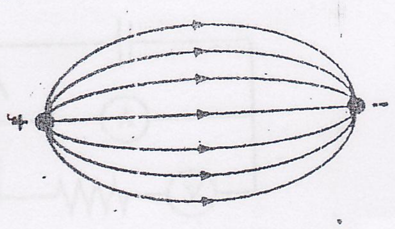
32. Which of the following would be involved in the production of fringes in a Young's double-slit experiment?

I. Diffraction
II. Refraction
III. Interference

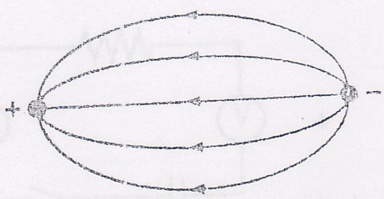
(A) I only
(B) III only
(C) I and III only
(D) I, II and III

Which of the following diagrams represents the electric field existing between two oppositely charged point charges?

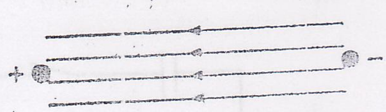
(A)



(B)



(C)



(D)

An electric current in a metal consists of a flow of

(A) neutrons
(B) protons
(C) electrons
(D) ions

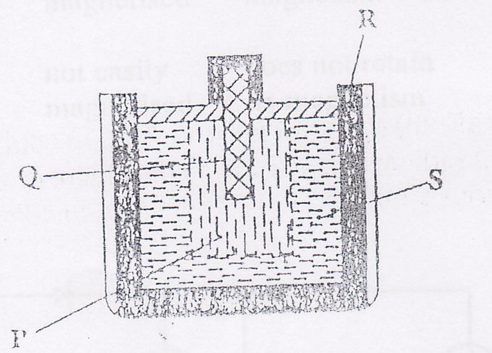
Which of the following appliances would require the fuse with the largest rating?

(A) A kettle marked 1 800 W 120 V
(B) A freezer marked 360 W 120 V
(C) An iron marked 1 200 W 240 V
(D) A stove marked 12 000 W 240 V

42. Which of the following formulae (in which the symbols have their usual meaning) is a mathematical expression of Ohm's Law?

(A) $V = IR$
(B) $P = VI$
(C) $Q = IT$
(D) $E = PT$

43. The diagram below shows the construction of a zinc-carbon cell.



Which of the following is correct?

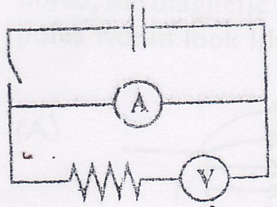
(A) Q is carbon, R is zinc.
(B) P is copper oxide, Q is carbon.
(C) R is iron and S is ammonium chloride.
(D) S is sulphuric acid and P is manganese dioxide.

44. Which of the following concerning ammeters is correct?

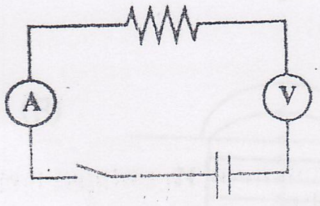
	Its resistance	Circuit Connection
(A)	Low	in series
(B)	Low	in parallel
(C)	High	in parallel
(D)	High	in series

A student requires a circuit to measure the resistance of a resistor. Which of the circuits below is correctly connected?

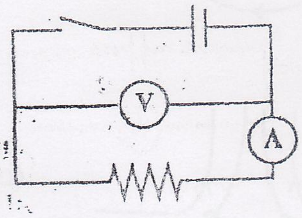
(A)



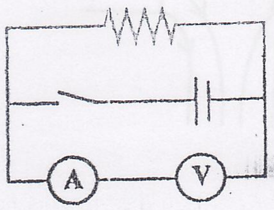
(B)



(C)



(D)



Which of the following relationships gives a correct value for the combined resistance of resistors R_1, R_2, and R_3 connected in parallel?

(A) $R_T = R_1 + R_2 + R_3$

(B) $R_T = \dfrac{R_1 R_2 R_3}{R_1 + R_2 + R_3}$

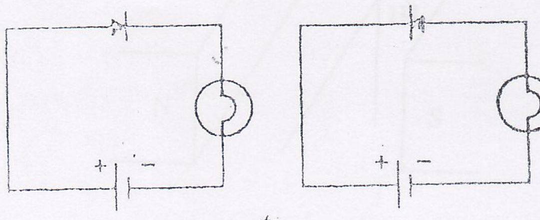
(C) $R_T = \dfrac{1}{R_1} + \dfrac{1}{R_2} + \dfrac{1}{R_3}$

(D) $\dfrac{1}{R_T} = \dfrac{1}{R_1} + \dfrac{1}{R_2} + \dfrac{1}{R_3}$

47. Which of the following pairs of statements is true for both iron and steel?

	Iron is	Steel
(A)	easily magnetised	does not retain magnetism
(B)	not easily magnetised	retains its magnetism well
(C)	easily magnetised	retains its magnetism well
(D)	not easily magnetised	does not retain its magnetism

48.



A experiment was conducted using the circuit diagrams shown above. The same components were used and the bulb was lit to normal brightness in each case.

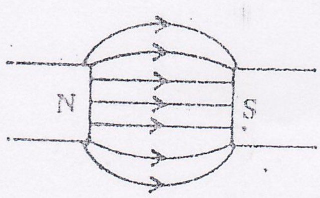
Which of the following statements would be correct?

I. The bulb is defective.
II. The battery is defective.
III. The diode is defective.

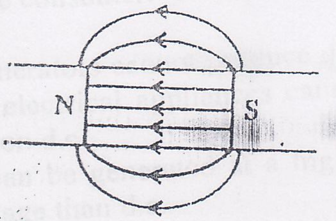
(A) I only
(B) III only
(C) I and II only
(D) II and III only

49. A north pole and a south pole of two equally strong magnets are brought near to each other. If the earth's magnetic field is ignored, the magnetic field lines between the poles would look like

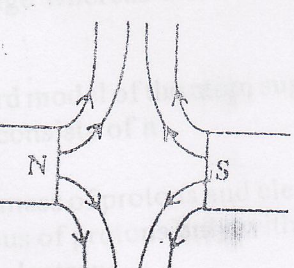
(A)



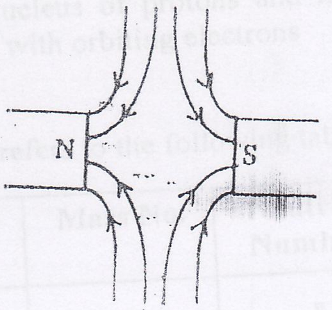
(B)



(C)



(D)

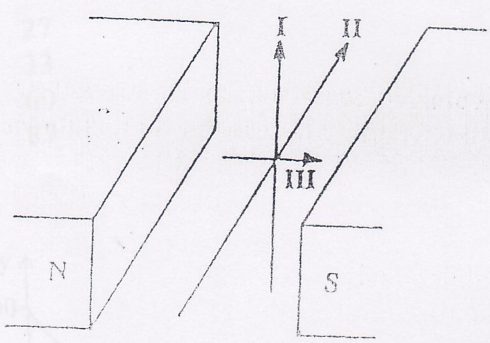


50. An electromagnet consists of insulated wire wrapped around an iron core.

It works because

(A) iron is a good electrical conductor
(B) a magnetic field is produced inside the coil
(C) an electric field is produced inside the coil
(D) iron is always magnetised

51.



The figure above shows two permanent magnets and three wires carrying current in the directions indicated. On which wire(s) will a force be exerted?

(A) I only
(B) II only
(C) I and II only
(D) I, II and III

52. Which of the following make use of the magnetic field generated by an electric current?

I. d.c. motor
II. Moving coil/loudspeaker
III. Electric iron

(A) I and II only
(B) I and III only
(C) II and III only
(D) I, II and III

A conductor, rotating in a uniform magnetic field, induces maximum instantaneous current when the conductor cuts the magnetic field lines at

(A) 30°
(B) 45°
(C) 90°
(D) 180°

An electricity company supplies a.c. rather than d.c. to the consumer. This is because

(A) its generators cannot produce d.c.
(B) most electrical appliances cannot run on d.c.
(C) a.c. can be generated at a higher voltage than d.c.
(D) a.c. can be stepped up to a higher voltage whereas d.c. cannot

The Rutherford model of the atom suggests that an atom consists of a

(A) solid mass of protons and electrons
(B) nucleus of protons only with orbiting electrons
(C) nucleus of equal numbers of neutrons and electrons with orbiting protons
(D) nucleus of protons and neutrons with orbiting electrons

Item 56 refers to the following table.

Nuclei	Mass No.	Neutron Number
P	16	8
Q	13	9
R	18	10
S	21	11

Which pair of nuclei are isotopes?

(A) P and Q
(B) Q and R

57. Which of the following electromagnetic radiations is produced only by a change in a nucleus?

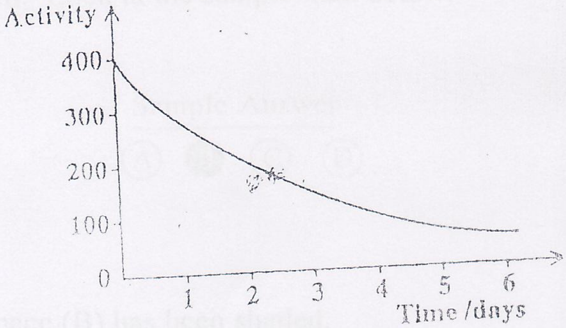
(A) Ultra-violet radiation
(B) Infra-red radiation
(C) Gamma radiation
(D) Radio waves

58. The element cobalt is represented by the expression $^{60}_{27}Co$.

This means that its neutron number is

(A) 27
(B) 33
(C) 60
(D) 87

59.



The activity of a radioactive substance was measured at suitable intervals over a period of days and its radioactive decay curve plotted. The half life is

(A) 1 day
(B) 2 days
(C) 3 days
(D) 4 days

60. Which of the following scientists discovered that the nucleus also contained neutrons?

(A) Niels Bohr
(B) James Chadwick
(C) Marie Curie
(D) Joseph Thompson